Master ELISE

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Digital Communications 1

Mahmoud Addouche

Lab Report — TP 1

Evaluating the Contribution of OFDM and OTFS

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Contents

1 Introduction

The reliability of a digital communication system depends heavily on how well the modulation scheme handles the distortions introduced by the transmission channel. In practice, radio channels are affected by multipath propagation and the Doppler effect, both of which degrade signal quality in ways that simple modulation techniques cannot easily handle.

This lab report covers a progressive study of three modulation approaches, each representing a step forward in dealing with channel impairments. We start with NRZ baseband signalling, which is the most basic form of binary transmission. We then move to OFDM, the multicarrier scheme that has dominated wireless standards for the past two decades. Finally, we study OTFS, a more recent technique that operates in the delay-Doppler domain and is designed specifically for high-mobility environments.

The goal is to understand, through simulation, the limitations of each scheme and the reasoning behind the transition from one to the next. All simulations are carried out in MATLAB using a set of provided functions. The multipath channel used in the later parts of the lab is defined by three propagation paths:

$$\tau = [0,\ 2,\ 2] \text{ samples}, \quad \nu = [0,\ -3,\ +3], \quad h = [1,\ 0.1,\ 0.15]$$

where τ are the path delays, ν the Doppler indices, and h the complex path gains.

2 NRZ Baseband Signalling

2.1 Time-Domain Signal and Power Spectral Density

A binary frame of $N = 32$ bits is encoded using Non-Return-to-Zero (NRZ) format. To simulate an analogue-like signal, each bit is spread over 6 samples (samplesPerBit = 6) at a sampling frequency of $F_e = 1000$ Hz. The resulting bit rate is $F_b = F_e/\text{samplesPerBit} = 167$ Hz.

The time-domain signal and its Power Spectral Density (PSD) are shown in Figure 1. The spectrum takes the shape of a sinc-squared envelope, which is typical of rectangular pulse shaping.

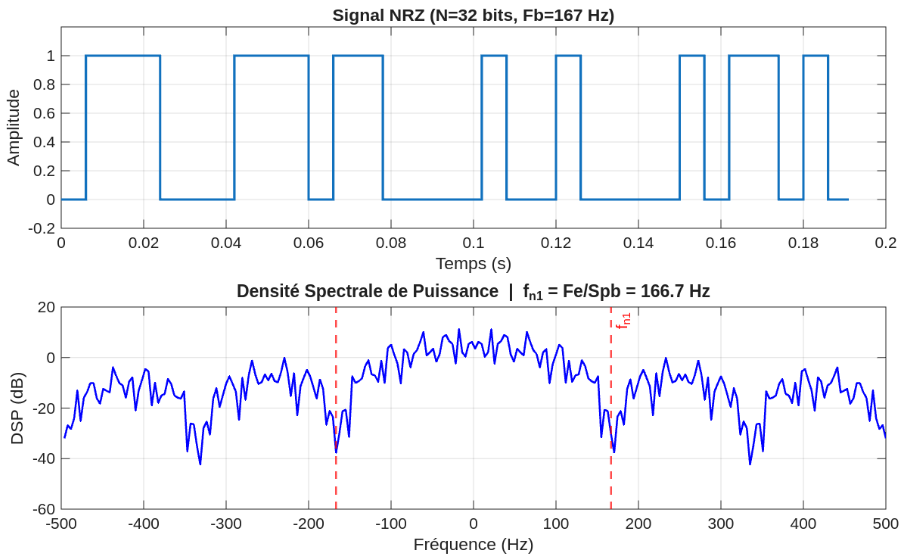

Figure 1: NRZ signal in the time domain (top) and its Power Spectral Density in dB (bottom). The red dashed lines indicate the first spectral null at $f_{n1} = \pm 166.7$ Hz.

2.2 First Spectral Null

The NRZ PSD follows a sinc-squared shape. Its first null corresponds to the inverse of the bit duration $T_b = \text{samplesPerBit}/F_e$, and is given by:

$$f_{n1} = \frac{F_e}{\text{samplesPerBit}} = \frac{1000}{6} \approx 166.7 \text{ Hz} \tag{1}$$

This frequency marks the boundary between the main spectral lobe and the first secondary lobe. Any channel that attenuates frequencies within the main lobe will affect all transmitted bits simultaneously, since each bit contributes to the entire spectrum.

2.3 Spectral Alteration and Signal Reconstruction

To illustrate the impact of frequency-selective fading, portions of the NRZ spectrum are set to zero and the signal is reconstructed via inverse FFT. Two scenarios are examined:

1. A small portion of the main lobe is removed (50 to 80 Hz).
2. The entire first secondary lobe is removed (167 to 333 Hz).

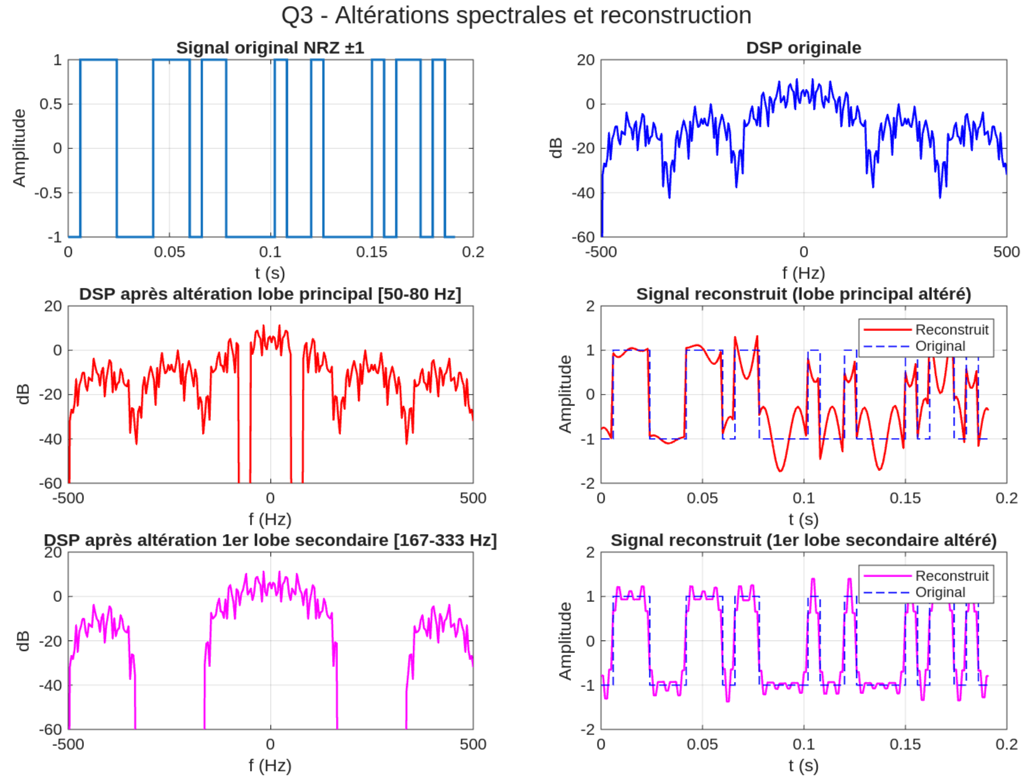

Figure 2: Spectral alterations and reconstructed signals. Top row: alteration of part of the main lobe and the resulting distorted signal. Bottom row: alteration of the first secondary lobe and the reconstructed signal.

Removing even a small portion of the main lobe causes heavy distortion across the entire reconstructed signal, with Gibbs-like oscillations appearing at every bit transition. By contrast, removing the first secondary lobe produces only minor ringing at the edges, since this lobe carries a small fraction of the total signal energy.

2.4 Effect of Frequency-Selective Fading on NRZ Symbols

Frequency-selective fading corrupts the entire transmitted frame, not just isolated symbols.

In NRZ encoding, each bit is represented by a rectangular pulse that occupies the full transmission bandwidth. Its spectral contribution is a sinc function spread across all frequency bins. When a frequency-selective fade attenuates a specific band, it removes spectral components that are shared by every bit in the frame. The damage cannot be isolated to a single symbol.

This was confirmed experimentally: zeroing a small portion of the main lobe (which carries energy from all bits equally) corrupts the entire reconstructed waveform. The conclusion is that NRZ is fundamentally ill-suited for frequency-selective channels, and this is the core motivation for moving to OFDM.

3 OFDM Multiplexing

Orthogonal Frequency Division Multiplexing splits a high-rate data stream into many lower-rate parallel streams, each modulated onto a separate orthogonal subcarrier. The time-domain signal is produced via the Inverse FFT. A cyclic prefix (CP) of duration T_g is added at the start of each symbol to absorb multipath echoes and prevent inter-symbol interference. The key property is that a frequency-selective fade now only affects a subset of subcarriers, leaving the others intact.

3.1 QAM-16 Constellation

With $L = 4$ bits per symbol, QAM-16 maps groups of 4 bits to one of 16 complex points arranged on a square grid. Gray coding ensures that adjacent symbols differ by only one bit, which minimises the number of bit errors when a symbol is misidentified as a neighbour. The constellation is implemented by the `myQAMmod` function, which also normalises the average symbol power to unity.

Figure 3 shows the 16 constellation points along with all 240 possible transitions between them.

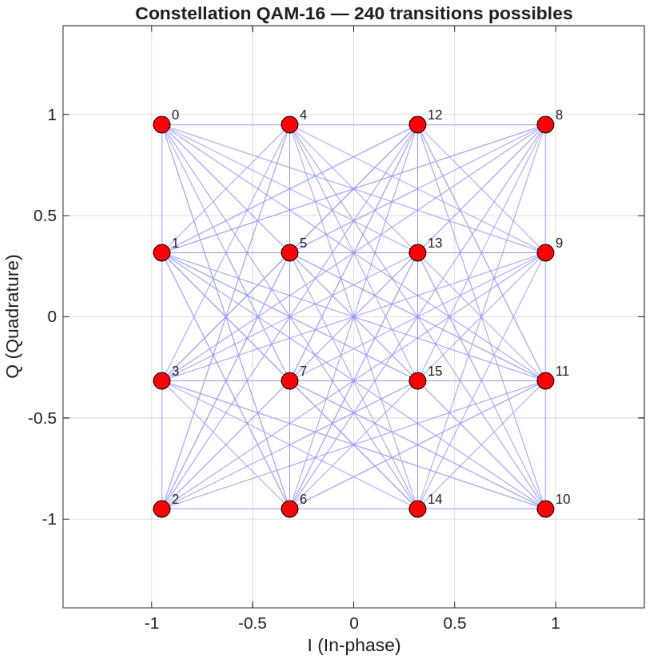

Figure 3: QAM-16 constellation. The 16 symbol points are shown in red. The blue lines represent all 240 possible transitions between consecutive symbols, illustrating the Gray-coded structure of the mapping.

3.2 OFDM Time-Domain Signal

The message "Master ELISE sem2§" (18 characters) is converted to QAM-16 symbols and transmitted as a single OFDM frame. With $L = 4$, the number of active subcarriers is $M = 18 \times 8/4 = 36$. The system operates at $F_s = 8000$ Hz with a useful symbol duration of $T_u = 8$ ms and a cyclic prefix of $T_g = 2$ ms, giving a spectral resolution of 64 bins, which comfortably accommodates the 36 active subcarriers.

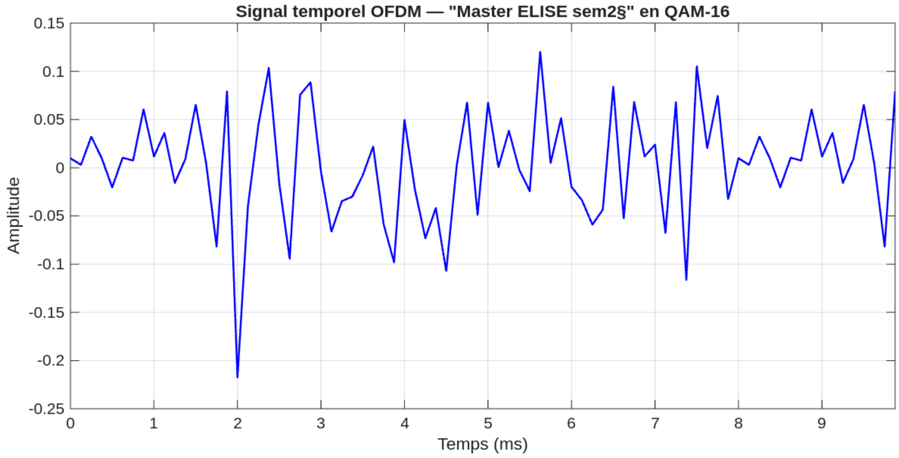

Figure 4: Time-domain OFDM signal for the transmitted message in QAM-16. The multi-frequency oscillation is the result of the superposition of 36 orthogonal subcarriers. Perfect reconstruction is verified at the receiver in the absence of noise.

3.3 Performance over an AWGN Channel

The failure rate is evaluated for 10 SNR values ranging from 10 to 20 dB. For each value, 20 independent trials are run. A trial is considered a failure if the decoded message differs from the transmitted one in any character.

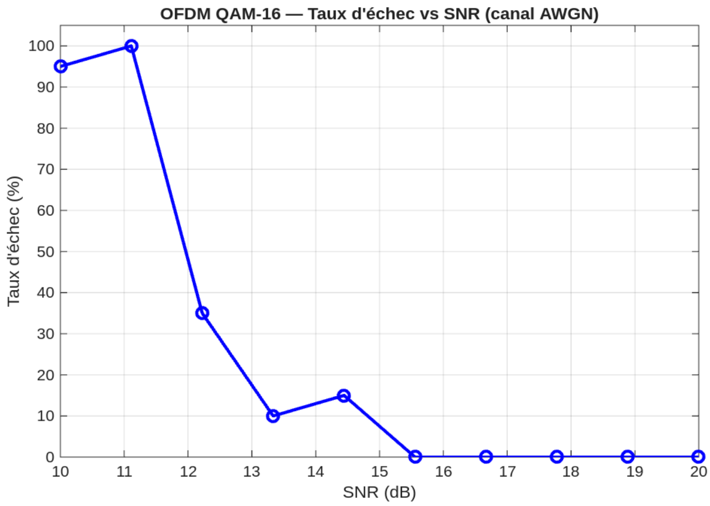

Figure 5: OFDM failure rate as a function of SNR over an AWGN channel. Each point is estimated from 20 independent trials. The failure rate transitions from near 100% at 10 dB to near zero above 15 dB.

The transition between high and low failure rates occurs around 13 to 15 dB. Below this threshold, the noise variance is large enough to push received symbols outside their correct decision regions. The slight non-monotonicity visible at 15.6 dB (1 failure out of 20) reflects the natural statistical variability of a small Monte Carlo sample.

3.4 COFDM with Hamming Error-Correcting Code

Coded OFDM (COFDM) adds redundancy before modulation using a Hamming(31, 26) block code. Each block of $k = 26$ data bits is encoded into $n = 31$ bits by appending $m = 5$ parity bits, giving a code capable of correcting any single-bit error per block. The message "Master1 ELISE" (13 characters, 104 bits, 4 blocks of 26 bits) is used for this comparison.

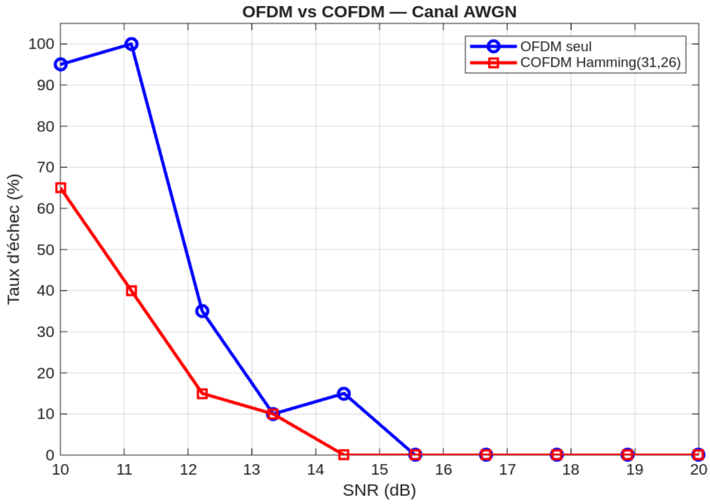

Figure 6: Failure rate comparison between OFDM alone (blue) and COFDM with Hamming(31, 26) coding (red) over an AWGN channel. The error-correcting code provides a gain of approximately 2 to 3 dB.

The Hamming code shifts the failure curve to lower SNR values by approximately 2 to 3 dB. At 10 dB, COFDM achieves a 70% failure rate compared to 95% for uncoded OFDM. This gain comes at the cost of a code rate of $k/n = 26/31 \approx 0.84$, meaning roughly 16% of the transmitted bits carry redundancy rather than information.

4 Multipath and Doppler Channel

4.1 Limitations of OFDM under Doppler Effect

OFDM relies on the assumption that the channel remains approximately constant over the duration of one OFDM symbol. In mobile environments, moving reflectors introduce Doppler frequency shifts. Each subcarrier centred at frequency $l \cdot \Delta f$ is shifted by the Doppler offset ν_d, which causes energy to leak into neighbouring subcarriers. This phenomenon is known as inter-carrier interference (ICI), and it destroys the orthogonality that OFDM depends on. Unlike AWGN, ICI is a systematic effect that does not diminish as transmit power increases.

4.2 Comparative Study in a Doppler Multipath Channel

The channel is characterised by the three paths defined in the introduction. The SNR is varied from 0 to 40 dB using 15 values, with 20 trials per point. Both uncoded OFDM and COFDM with Hamming(31, 26) are evaluated.

Path	Delay (samples)	Complex gain	Doppler index
1	0	1.00	0
2	2	0.10	-3
3	2	0.15	$+3$

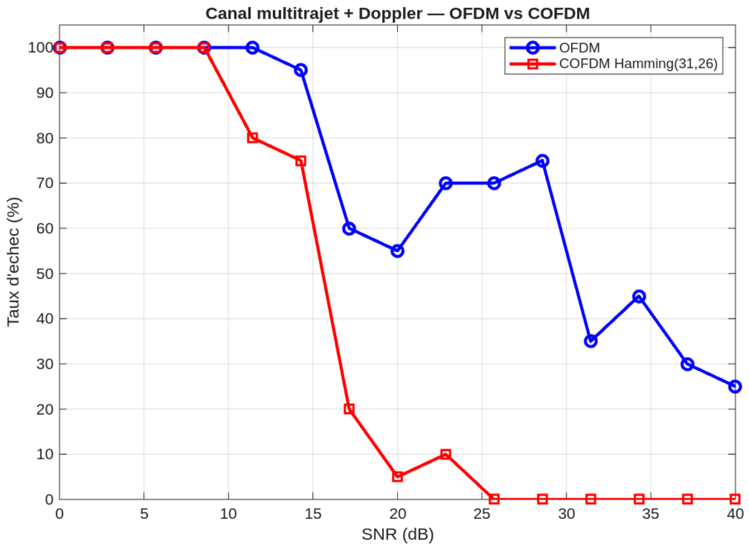

Figure 7: Failure rate in the multipath Doppler channel. OFDM (blue) stabilises around 50% failure even at 40 dB, demonstrating that ICI is irreducible regardless of SNR. COFDM (red) benefits from error correction at moderate SNR values but also reaches a performance floor at high SNR due to the systematic nature of ICI.

The result is striking: OFDM maintains roughly 50% failure at 40 dB, a level that would be unimaginable in a pure AWGN channel where QAM-16 achieves near-zero errors above 15 dB. The Hamming code helps at intermediate SNR values but cannot correct the systematic distortion caused by ICI at high SNR. This confirms that OFDM is fundamentally unsuitable for high-mobility channels, and points to the need for a different approach.

5 OTFS Modulation

5.1 Principle

OTFS (Orthogonal Time Frequency Space) encodes data in the delay-Doppler domain rather than the time-frequency domain. The key insight is that a multipath Doppler channel, which appears as a complex time-varying matrix in the frequency domain, becomes a sparse and stable two-dimensional convolution in the delay-Doppler domain. Each scatterer contributes a single energy peak at its delay and Doppler coordinates, regardless of how long the transmission lasts.

The processing chain is as follows. At the transmitter, the data grid $X_{dd}[m, n]$ in the delay-Doppler domain is converted to the time-frequency domain via the Inverse Symplectic Finite Fourier Transform (ISFFT), then passed through the standard OFDM modulator. At the receiver, the standard OFDM demodulator is applied first, followed by the Symplectic FFT (SFFT) to return to the delay-Doppler domain:

$$X_{dd} \xrightarrow{\text{ISFFT}} X_{tf} \xrightarrow{\text{OFDM mod}} x(t) \xrightarrow{\text{channel}} y(t) \xrightarrow{\text{OFDM demod}} Y_{tf} \xrightarrow{\text{SFFT}} Y_{dd}$$

The ISFFT is defined by:

$$X_{tf}[l,k] = \frac{1}{\sqrt{NM}} \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} X_{dd}[m,n] \exp\left(j2\pi\left(\frac{nk}{N} - \frac{ml}{M}\right)\right) \tag{2}$$

Its implementation combines a row-wise IFFT with a column-wise FFT, scaled by $\sqrt{N/M}$.

5.2 Validation Without Channel

To verify the OTFS implementation, 33 French verbs of exactly 15 characters each are transmitted through the OTFS chain with AWGN only at SNR = 40 dB. With $L = 4$ bits per symbol, each word produces $M = 15 \times 8/4 = 30$ symbols, giving a delay-Doppler grid of dimensions 30×33.

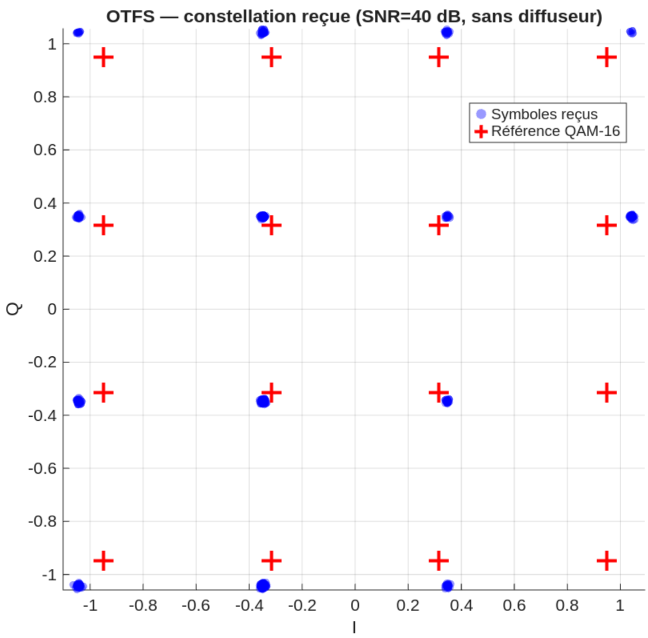

Figure 8: Received QAM-16 constellation after OTFS demodulation with no channel at SNR = 40 dB. The received symbols (blue dots) cluster tightly around the 16 reference points (red crosses). The result is 0 errors out of 33 words.

The result of 0 errors out of 33 words confirms that the ISFFT and SFFT are correctly implemented and that the pipeline is consistent with the theoretical OTFS chain.

5.3 Channel Estimation via Pilot Signal

To identify the channel parameters, a unit impulse is placed at position $(0, 0)$ of the delay-Doppler grid and transmitted through the multipath Doppler channel at SNR = -10 dB.

At the receiver, the channel response appears as energy peaks at the delay and Doppler coordinates of each scatterer.

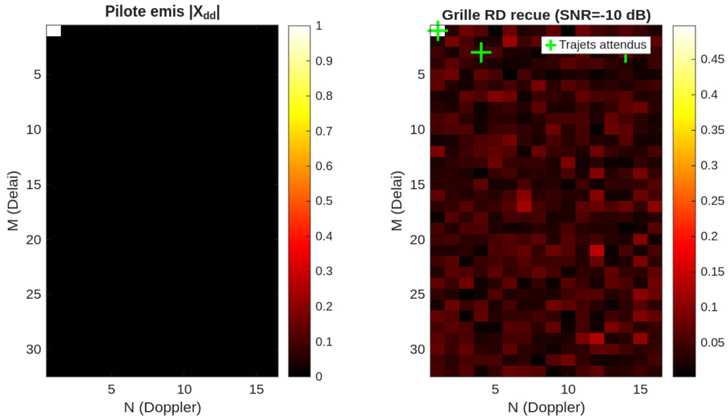

Figure 9: Left: the transmitted pilot impulse in the delay-Doppler domain. Right: the received delay-Doppler grid at SNR = −10 dB. Green crosses mark the positions of the three expected scatterers. At this very low SNR, only the dominant path is clearly visible above the noise floor.

At SNR $= -10$ dB, the noise floor masks the two secondary paths whose gains (0.10 and 0.15) are comparable to the noise level. Only the dominant path (delay $= 0$, Doppler $= 0$) is unambiguously identified. The three largest peaks detected are summarised below:

Peak	Delay (samples)	Doppler index	Estimated gain
1	0	0	0.517
2	1	+3	0.132
3	14	−6	0.123

This result illustrates a known limitation of pilot-based estimation: when secondary paths are weak relative to the noise, they cannot be resolved reliably. For the equalization step, the true channel parameters are used in order to evaluate the equalizer performance independently of the estimation accuracy.

5.4 Equalization and Constellation Comparison

With the channel known, equalization is performed using a two-dimensional Wiener filter in the delay-Doppler domain. The OTFS channel acts as a two-dimensional circular convolution between the transmitted grid and the channel point spread function (PSF). The PSF is obtained by passing a noiseless pilot impulse through the channel. The Wiener filter then inverts this convolution while regularising against noise:

$$\hat{X}_{dd}(\omega) = \frac{H^*(\omega)}{|H(\omega)|^2 + n_{\text{level}}} \cdot Y_{dd}(\omega) \tag{3}$$

where $H(\omega)$ is the two-dimensional Fourier transform of the PSF, $Y_{dd}(\omega)$ is the Fourier transform of the received delay-Doppler grid, and $n_{\text{level}} = 10^{-\text{SNR}/10}$ is the regularisation term.

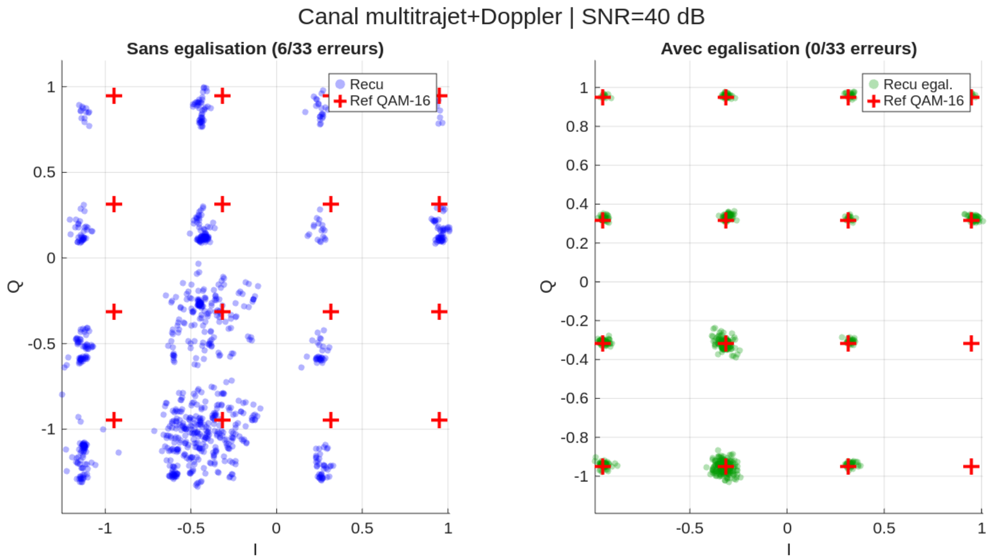

Figure 10: QAM-16 constellations in the multipath Doppler channel at SNR = 40 dB. Left: without equalization, the 16 symbol clusters are spread and overlapping due to inter-symbol and inter-carrier interference. Right: after two-dimensional Wiener equalization, the clusters are sharply separated and the result is 0 errors out of 33 words.

The equalization reduces the word error rate from approximately 5 to 6 out of 33 down to 0 out of 33. This result demonstrates the core advantage of OTFS: by representing the channel as a sparse, stable convolution in the delay-Doppler domain, a simple two-dimensional Wiener filter is sufficient to undo the effects of multipath propagation and Doppler simultaneously.

6 Conclusion

This lab has examined three digital modulation techniques across a range of increasingly challenging channel conditions.

NRZ baseband signalling is the simplest approach but the most fragile. Because each bit occupies the full signal bandwidth, a frequency-selective fade corrupts all symbols simultaneously. There is no spectral isolation between symbols, and no straightforward way to recover from band-limited distortion.

OFDM solves this problem by splitting the signal across many narrow subcarriers. A selective fade only affects a subset of them, and error-correcting codes such as Hamming can recover the lost information. Over an AWGN channel, OFDM with QAM-16 requires an SNR of approximately 14 to 15 dB to achieve reliable transmission, and COFDM reduces this threshold by 2 to 3 dB. However, OFDM relies on the channel being stable across one symbol duration. In mobile environments, Doppler shifts create inter-carrier interference that is systematic and irreducible: even at 40 dB SNR, the failure rate remains around 50%, making OFDM unsuitable for high-mobility scenarios.

OTFS addresses this by working in the delay-Doppler domain, where each scatterer leaves a localised, stable imprint regardless of mobility. The two-dimensional Wiener equalizer exploits this structure to fully recover the transmitted symbols. The simulation achieves

0 errors out of 33 words in a three-path Doppler channel at 40 dB, compared to approximately 15% failure for OFDM under the same conditions.

The progression from NRZ to OFDM to OTFS reflects a broader principle in communications engineering: the most effective modulation scheme is one whose signal representation matches the structure of the channel. OTFS, by working in the natural domain of mobile wireless channels, is a strong candidate for future high-mobility systems such as satellite communications and vehicle-to-everything networks.

A Code Summary

File	Description
Q2_NRZ.m	NRZ signal generation, time-domain plot, PSD computation and marking of the first spectral null.
Q3_Alteration.m	Spectral zeroing of main and secondary lobes via FFT, followed by IFFT reconstruction.
Q4A_Constellation.m	QAM-16 constellation display with all 240 inter-symbol transitions.
Q4B_OFDM_signal.m	OFDM modulation of a text message and time-domain signal plot.
Q4C_BER_AWGN.m	Monte Carlo failure rate simulation over AWGN channel, SNR from 10 to 20 dB.
Q4D_COFDM.m	COFDM with Hamming(31, 26); failure rate comparison with uncoded OFDM.
Q5_Doppler.m	Failure rate study for OFDM and COFDM in the multipath Doppler channel.
Q6A_OTFS_clean.m	OTFS modulation and demodulation of 33 words without a propagation channel.
Q6B_PilotEstimation.m	Delay-Doppler domain channel estimation using a pilot impulse at SNR = -10 dB.
Q6C_Equalisation.m	Two-dimensional Wiener equalization in the delay-Doppler domain; constellation comparison.

B System Parameters

Parameter	Value	Description
F_e	1000 Hz	NRZ sampling frequency
samplesPerBit	6	NRZ oversampling factor
N	32 bits	NRZ frame length
L	4 bits/symbol	QAM modulation order
F_s	8000 Hz	OFDM and OTFS sampling frequency
T_u	8 ms	Useful OFDM symbol duration
T_g	2 ms	Cyclic prefix duration
M (OFDM)	36	Number of active subcarriers
M (OTFS)	30	Number of delay-domain bins
N (OTFS)	33	Number of Doppler-domain bins
n	31	Hamming codeword length
k	26	Hamming data bits per block